



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



Risk-Sensitive Portfolio Management with Multi-Agent Reinforcement Learning for Portfolio Optimization

Submitted in partial fulfilment of the requirements of the degree of
Bachelor of Technology in Computer Science and Engineering (Data Science)

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A.Y. 2025-26

Certificate

This is to certify that the project entitled, **“Risk-Sensitive Portfolio Management with Multi-Agent Reinforcement Learning for Portfolio Optimization”** is a bonafide work of **“Harshil Solanki”** (60009220166), **“Heet Choudhari”** (60009220129), and **“Dhruv Shah”** (60009220132) submitted in partial fulfillment of the requirement for the award of the Bachelor of Technology in Computer Science and Engineering(Data Science).

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We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all the principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will cause disciplinary action by the Institute and can also evoke penal action from the sources, which have thus not been properly cited or from whom proper permission has not been taken when needed.

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This project report entitled (**Risk-Sensitive Portfolio Management with Multi-Agent Reinforcement Learning for Portfolio Optimization**) by (**Harshil Solanki, Heet Choudhari and Dhruv Shah**) is approved for the degree of B.Tech. in Computer Science and Engineering (Data Science).

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to everyone who has contributed to the successful completion of our research project, "Risk-Sensitive Portfolio Management using Multi-Agent Reinforcement Learning for Portfolio Optimization." First and foremost, we would like to extend our heartfelt thanks to Prof. Ruchika Patil, our project guide, for her invaluable guidance, continuous encouragement, and insightful feedback throughout the course of this work. Her expertise in the field and her constant motivation helped us refine our research direction and overcome the various challenges encountered during the project development phase.

We would also like to express our appreciation to Dr. Kriti Srivastava, Head of the Department of Computer Science and Engineering (Data Science), for her consistent support, encouragement, and for providing us with the necessary resources and a conducive research environment. Our gratitude also extends to all the faculty members of the Department of Computer Science and Engineering (Data Science) at Dwarkadas J. Sanghvi College of Engineering, for their valuable inputs, assistance, and continuous academic support that enriched our learning experience.

Finally, we are thankful to Dr. Hari Vasudevan, Principal of Dwarkadas J. Sanghvi College of Engineering, for providing access to institutional resources, research facilities, and an ecosystem that promotes learning, innovation, and practical exploration in the field of Data Science. We deeply appreciate the time, effort, and encouragement provided by all those who contributed directly or indirectly to the successful completion of this project.

Abstract

Traditional portfolio optimization frameworks such as Markowitz's Mean-Variance theory and its subsequent extensions have long been the cornerstone of modern finance. However, these methods assume a stationary distribution of returns and linear correlations, which fail to capture the dynamic, non-linear, and fat-tailed behavior of real-world financial markets. This results in significant underperformance and vulnerability during periods of high volatility and systemic shocks. To address these limitations, our project introduces a Risk-Sensitive Portfolio Management system based on Hierarchical Multi-Agent Reinforcement Learning (MARL), integrating risk-awareness and adaptability into the core of the decision-making process.

In this framework, each asset is modeled as an autonomous Actor Agent that interacts cooperatively with other agents within a shared environment. These agents are guided by a Graph Neural Network (GNN) encoder that continuously learns inter-asset relationships and market dependencies, capturing dynamic correlations and contagion effects that arise during market turbulence. The system employs a Centralized Distributional Critic that optimizes beyond traditional mean-return objectives by incorporating distributional risk measures such as Conditional Value at Risk (CVaR) and Maximum Drawdown (MDD). This allows the model to explicitly account for tail risks and uncertainty, producing more resilient allocation strategies.

A Meta-Controller layer operates on top of this structure, adaptively adjusting multi-objective reward weights, rebalancing frequencies, and transaction constraints to align with shifting market regimes. This hierarchical control not only enhances responsiveness but also ensures that the learned policies maintain long-term robustness under transaction costs, slippage, and liquidity frictions—factors often overlooked in academic studies but critical in real-world deployment.

Through extensive simulated experiments and backtesting on historical financial data, the proposed system demonstrates significant improvements in risk-adjusted performance metrics such as the Sharpe Ratio, Sortino Ratio, and Calmar Ratio, outperforming baseline single-agent and traditional optimization models. The project highlights the transformative potential of reinforcement learning in financial decision systems, offering a scalable and intelligent framework for adaptive, risk-sensitive asset management.

Keywords: Multi-Agent Reinforcement Learning (MARL), Portfolio Optimization, Risk-Sensitive Learning, Conditional Value at Risk (CVaR), Graph Neural Networks (GNN), Centralized Critic, Hierarchical Policy Learning, Dynamic Asset Allocation, Financial Risk Management, Reinforcement Learning in Finance.

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Chapter 1

Introduction

This chapter provides an overview of the research problem, explaining the motivation behind developing a risk-sensitive, adaptive portfolio management system using Multi-Agent Reinforcement Learning (MARL). It outlines the background, objectives, and the need for intelligent, data-driven approaches to achieve robust and risk-aware financial decision-making in dynamic market environments.

1.1 Introduction

The field of portfolio management has witnessed a paradigm shift with the growing integration of artificial intelligence and machine learning techniques into financial decision-making. Traditional investment strategies, grounded in static mathematical models such as Mean-Variance Optimization, often fall short in handling the dynamic and uncertain nature of modern financial markets. These conventional methods rely heavily on assumptions of market efficiency, normally distributed returns, and fixed correlations among assets—assumptions that rarely hold true in real-world trading environments.

In recent years, Reinforcement Learning (RL) has emerged as a promising approach for sequential decision-making in complex and uncertain domains. By continuously interacting with an environment and learning from feedback in the form of rewards, RL models can discover adaptive strategies that evolve with changing market conditions. However, most existing RL-based financial models treat the portfolio as a single decision-making entity, which limits their ability to capture interactions among multiple assets and their evolving dependencies.

This limitation has motivated the development of Multi-Agent Reinforcement Learning (MARL), where multiple agents operate collaboratively or competitively to achieve a shared financial objective. In our research project, we extend this

idea to construct a Risk-Sensitive Multi-Agent framework for portfolio optimization. The proposed system leverages advanced neural architectures such as Graph Neural Networks (GNNs) to model inter-asset relationships and employs a Distributional Critic to quantify and control downside risk through Conditional Value at Risk (CVaR) and Maximum Drawdown (MDD) optimization.

The primary goal of this research is to design a dynamic and adaptive trading policy capable of balancing profitability with risk management, enabling robust performance during volatile market conditions. This introduction sets the foundation for understanding the motivation, challenges, and overall objectives of our work in building a scalable, intelligent, and risk-aware financial decision-making system using Multi-Agent Reinforcement Learning.

1.2 Project Overview

This project, titled “Risk-Sensitive Portfolio Management with Multi-Agent Reinforcement Learning for Portfolio Optimization,” aims to develop an adaptive and risk-aware system for intelligent financial decision-making. Traditional portfolio optimization methods, such as Mean-Variance theory, struggle to handle the dynamic, uncertain, and interdependent nature of modern markets. To overcome these limitations, our approach leverages Multi-Agent Reinforcement Learning (MARL), where each asset is modeled as an autonomous agent that cooperatively learns optimal allocation strategies.

The framework integrates a Graph Neural Network (GNN) to capture dynamic relationships between assets and a Centralized Distributional Critic to optimize both returns and risk measures like Conditional Value at Risk (CVaR) and Maximum Drawdown (MDD). A Meta-Controller further enhances adaptability by adjusting reward weights and rebalancing frequency based on market conditions.

By simulating real-world trading factors such as transaction costs and slippage, the system aims to achieve robust performance and improved risk-adjusted returns compared to traditional models. In essence, this project bridges reinforcement learning and quantitative finance to create a dynamic, risk-sensitive, and data-driven portfolio management framework.

1.3 Motivation

Financial markets are highly dynamic, uncertain, and influenced by numerous interdependent factors such as macroeconomic events, investor sentiment, and global

liquidity flows. Traditional portfolio optimization methods like the Mean-Variance model assume stable market conditions and fixed asset correlations, which often fail during periods of high volatility or systemic shocks. These limitations highlight the need for a more adaptive, intelligent, and risk-aware approach to portfolio management.

Recent advancements in Reinforcement Learning (RL) and Deep Learning have opened new possibilities for building systems that can learn optimal trading and allocation strategies directly from market interactions. However, single-agent RL models struggle to capture the cooperative and competitive dynamics that naturally exist between assets in a portfolio. This gap motivates the use of Multi-Agent Reinforcement Learning (MARL), where multiple agents work together to optimize the overall portfolio's performance while managing individual asset-level risks.

The motivation behind this project is to design a framework that not only seeks high returns but also emphasizes stability and resilience against market uncertainty. By integrating MARL with risk-sensitive objectives such as Conditional Value at Risk (CVaR) and Maximum Drawdown (MDD), the proposed system aims to deliver consistent, risk-adjusted performance. Ultimately, this research aspires to demonstrate how intelligent, data-driven agents can revolutionize financial decision-making by enabling adaptive and robust portfolio management strategies in real-world markets.

1.4 Project Outcome

The primary outcome of this project is the development of a Risk-Sensitive Multi-Agent Reinforcement Learning (MARL) framework capable of adaptive and intelligent portfolio management under real-world financial constraints. The system successfully integrates Graph Neural Networks (GNNs) for dynamic correlation modeling, a Centralized Distributional Critic for risk-aware optimization, and a Meta-Controller for adaptive policy adjustment based on market regimes.

Through extensive simulation and backtesting on historical market data, the model demonstrates enhanced risk-adjusted returns, improved portfolio stability, and superior performance metrics such as Sharpe Ratio, Sortino Ratio, and Calmar Ratio, when compared to traditional and single-agent reinforcement learning methods. The outcome validates that multi-agent cooperation combined with distributional and hierarchical learning significantly improves portfolio resilience during volatile market conditions.

Overall, the project provides a scalable and practical foundation for deploying

AI-driven portfolio management systems capable of learning from market behavior, adapting dynamically, and optimizing both performance and risk exposure in a unified framework.

1.5 Organization of the Report

This report is organized into six chapters, each providing a detailed and systematic understanding of the research work carried out in this project. Chapter 1 introduces the overall concept of the project, including its background, motivation, objectives, and expected outcomes. Chapter 2 presents a comprehensive literature review, summarizing existing research in portfolio optimization, reinforcement learning applications in finance, and identifying the research gap that this project aims to address. Chapter 3 defines the problem statement and outlines the specific objectives that guide the development of the proposed system. Chapter 4 describes the proposed methodology in detail, explaining the architecture of the Multi-Agent Reinforcement Learning framework, the use of Graph Neural Networks, and the role of risk-sensitive optimization techniques. Chapter 5 focuses on the dataset used, including data collection, preprocessing steps, and simulation setup for model training and evaluation. Finally, Chapter 6 concludes the report by summarizing the results, discussing key findings, identifying limitations, and highlighting the potential future scope for enhancing and extending the project. This organization ensures a coherent flow from conceptualization to implementation, providing a complete overview of the research process and outcomes.

Chapter 2

Literature Review

This chapter presents a comprehensive review of existing research in portfolio optimization using Reinforcement Learning (RL) and Multi-Agent Reinforcement Learning (MARL). It analyzes prior approaches, identifies their limitations in handling tail-risk, cross-asset dependencies, and dynamic market regimes, and establishes the foundation for our proposed risk-sensitive MARL framework that integrates graph-based modeling, distributional critics, and hierarchical policy control to achieve adaptive and robust portfolio management.

2.1 Preliminaries

Portfolio optimization is a core problem in quantitative finance that aims to allocate assets in a way that maximizes returns while minimizing risk. Traditional models, such as Markowitz’s Mean-Variance Theory (MPT), assume that returns are normally distributed and that asset correlations remain stable over time. However, in real-world markets, asset returns exhibit non-stationary, non-linear, and fat-tailed behavior, making these assumptions unrealistic. This has led to the growing adoption of Reinforcement Learning (RL) and, more recently, Multi-Agent Reinforcement Learning (MARL) for financial decision-making.

Reinforcement Learning (RL) is a branch of machine learning where an agent interacts with an environment to learn optimal actions through trial and error, guided by rewards. The agent’s goal is to learn a policy that maximizes cumulative rewards over time. Algorithms such as Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), and Deep Deterministic Policy Gradient (DDPG) have shown success in continuous control problems, including financial trading. However, these single-agent frameworks often fall short in multi-asset environments where interactions among assets significantly impact performance.

Multi-Agent Reinforcement Learning (MARL) extends RL by allowing multiple agents to interact and learn simultaneously in a shared environment. Each agent represents an entity—such as an asset or trading strategy—and learns to cooperate or compete with others to achieve a common objective. This framework is particularly suited for portfolio management, where interdependencies among assets, market sectors, and trading actions influence the overall portfolio outcome.

In our proposed research, the MARL framework is enhanced with Graph Neural Networks (GNNs) to model dynamic asset correlations and Distributional Critics to capture full return distributions, allowing optimization over tail-risk measures such as Conditional Value at Risk (CVaR) and Maximum Drawdown (MDD). A Meta-Controller governs adaptive risk control by adjusting reward weights and rebalancing frequency based on market regimes. Together, these components provide a foundation for a risk-sensitive, adaptive, and intelligent portfolio optimization system capable of real-world deployment.

2.2 Literature Review

The development of autonomous and adaptive portfolio management systems has gained significant momentum with the rise of Reinforcement Learning (RL) and Multi-Agent Reinforcement Learning (MARL). Traditional approaches based on Modern Portfolio Theory (MPT) and mean-variance optimization often fail to adapt to the dynamic, non-stationary nature of financial markets, leading to poor generalization during volatile periods. Recent studies have applied Deep Reinforcement Learning (DRL) algorithms such as PPO, DDPG, and SAC to portfolio optimization, showing improved returns compared to conventional models. However, these methods largely remain risk-neutral and struggle to handle extreme market fluctuations or systemic shocks.

MARL has emerged as a powerful paradigm for addressing the multi-dimensional challenges of financial decision-making. By modeling assets or strategies as autonomous agents that interact within a shared market environment, MARL enables cooperative learning and better representation of inter-asset dependencies. Research has shown that this approach can lead to more stable and diversified portfolios, but key challenges still remain—particularly in modeling cross-asset relationships, managing tail-risk, and maintaining adaptability under changing market conditions. To address these issues, our proposed framework integrates Graph Neural Networks (GNNs), Distributional Critics, and Hierarchical Control to capture systemic dependencies, optimize Conditional Value at Risk (CVaR), and dynamically rebalance

portfolios based on market regimes.

2.2.1 Similar Applications

Several studies have explored the use of RL and MARL in financial portfolio management. For instance, Chorasiya and Kingler (2025) applied single-agent DRL models such as PPO and DDPG for Indian stock markets, demonstrating improved cumulative returns but lacking tail-risk awareness and adaptability. The MARS framework by Jiayi Chen (2025) introduced a hierarchical MARL structure with meta-controllers and heterogeneous agents, achieving strong risk control yet limited scalability. Zhang et al. (2023) combined MADDPG with rule-based strategies for regime-adaptive trading, showing enhanced performance but static reward design.

Other frameworks such as RiskQ (2023) and MASA (Li et al., 2024) have incorporated risk-sensitive elements and agent collaboration, proving the benefits of distributional critics and adaptive coordination. However, they often suffer from restricted data universes, single-utility optimization, or lack of real-world frictions like transaction costs and liquidity impact. These studies highlight the ongoing transition from isolated DRL models toward multi-agent, risk-aware systems that better emulate real market dynamics.

2.2.2 Related Research

Recent literature has shifted focus from return maximization toward risk-sensitive and adaptive learning in financial decision-making. Wang and Liu (2025) proposed a volatility-aware DRL model using transformer-based feature extraction and multi-agent ensembles, achieving improved Sharpe and Sortino ratios across volatile periods. Huang and Tanaka (2022) introduced MSPM, a modular multi-agent RL system that combines per-asset evolving agents with a strategic portfolio allocator, emphasizing scalability and information sharing. Similarly, Al Maruf et al. (2023) developed a risk-aware distributed MARL algorithm using CVaR Q-learning, proving convergence and improved coordination among agents.

While these methods mark substantial progress, they typically lack dynamic control mechanisms capable of adjusting to varying market conditions. Our research builds upon these foundations by introducing a Hierarchical Multi-Agent RL framework that unifies dynamic risk optimization, cross-asset graph modeling, and adaptive meta-control. This approach bridges the gap between theoretical reinforcement learning advancements and their practical, real-world application in risk-sensitive portfolio management.

2.3 Gap Analysis

The review of existing literature reveals that while Reinforcement Learning (RL) and Multi-Agent Reinforcement Learning (MARL) have made significant progress in financial portfolio optimization, several critical gaps remain that hinder their practical effectiveness in real-world markets. Most conventional RL models focus on maximizing expected mean returns, inherently ignoring low-probability, high-impact tail events that dominate financial crises. This risk-neutral behavior leads to poor performance under heavy-tailed market distributions, where extreme drawdowns and volatility spikes are far more frequent than predicted by Gaussian assumptions. Traditional models fail to account for leptokurtosis and the asymmetric distribution of returns, resulting in severe capital loss during systemic shocks.

Another major limitation lies in the inadequate modeling of cross-asset dependencies. Many MARL systems train agents using isolated asset-level observations, leading to “local myopia” where agents make decisions without understanding the broader market context. This omission causes redundant risk exposures and poor diversification, as correlated assets may move in tandem during crises, amplifying portfolio losses. Without a mechanism to capture inter-asset relationships, these systems remain vulnerable to systemic shocks that undermine portfolio stability.

Furthermore, most existing frameworks employ single-objective reward functions, prioritizing return maximization while ignoring conflicting real-world objectives such as minimizing risk, controlling drawdowns, and reducing transaction costs. This simplification leads to unrealistic trading policies with excessive turnover, which are impractical in live trading environments. Real-world portfolio management requires balancing multiple objectives dynamically, something that static reward-weighting schemes cannot achieve.

Lastly, current reinforcement learning approaches suffer from limited feedback integration and short-term myopia. Agents typically rely on immediate step-wise rewards, which fail to reflect long-term performance measures such as the Sharpe Ratio or Maximum Drawdown. This makes them unsuitable for strategic, long-horizon investment planning.

In summary, the key research gaps identified are: the absence of tail-risk optimization, poor cross-asset dependency modeling, oversimplified single-objective reward design, and lack of long-term feedback integration. Addressing these gaps, our proposed Risk-Sensitive Hierarchical Multi-Agent Reinforcement Learning framework introduces a Centralized Distributional Critic for full return distribution modeling, a Graph Neural Network (GNN) encoder for dynamic inter-asset relationships, and a Meta-Controller for adaptive multi-objective reward balancing. Together,

these innovations aim to build a robust, risk-aware, and adaptive portfolio management system capable of outperforming static and risk-neutral benchmarks in dynamic market environments.

2.4 Summary

This chapter reviewed existing research on Reinforcement Learning (RL) and Multi-Agent Reinforcement Learning (MARL) for portfolio optimization. Traditional models like Mean-Variance Theory and basic DRL frameworks improved returns but failed to address non-stationary markets, tail-risk, and inter-asset dependencies. While MARL-based methods introduced cooperation and adaptability, most lacked scalability, real-world testing, and explicit risk-sensitive optimization.

The identified gaps—such as poor modeling of cross-asset relationships, limited dynamic risk control, and static reward structures—highlight the need for our proposed Risk-Sensitive Hierarchical MARL framework, which integrates GNNs, Distributional Critics, and a Meta-Controller to achieve adaptive, risk-aware, and robust portfolio management.

Table 2.1: Survey analysis table.

Sr.no.	Name of the Paper	Methods used	Results	Gaps
1	Chorasiya & Kinger (2025)[1]	Single-agent DRL (PPO, DDPG) with explainability	Sharpe 1.39, Ret 34%	No tail-risk, static, lacks adaptability
2	Jiayi Chen (MARS, 2025)[2]	Hierarchical MARL + Meta-Controller	Sharpe 2.84, MDD -5.39	Limited scalability, weak inter-links
3	Zhang et al. (2023)[3]	Hybrid MADDPG + rule-based overlays	Sharpe 2.09, Ret 9.6%	Static reward, no regime adaptivity
4	RiskQ (2023)[4]	Distributional MARL, quantile critic	Risk-aware coordination proven	Theoretical only, no financial test
5	Vadori et al. (2022)[5]	PPO-based multi-agent simulator	Realistic trading behavior	No tail-risk, high compute cost
6	Li et al. (2024)[6]	Self-adaptive MARL (MASA)	Better AR 8.8%, MDD 31%	No CVaR, small dataset
7	Wang & Liu (2025)[7]	DRL ensemble, volatility-aware reward	Better Sharpe & Sortino	No liquidity, limited validation
8	Choudhary et al. (2025)[8]	Multi-reward DRL (return, risk, MDD)	Outperformed baselines	Stock-only, fixed portfolio
9	Huang & Tanaka (2022)[9]	Modular MARL (EAM + SAM)	80% win vs. baselines	No slip-page/liquidity modeling
10	Al Maruf et al. (2023)[10]	Distributed CVaR Q-learning	Agents reach value consensus	Limited to discrete spaces
11	Hao et al. (2025)[11]	Cooperative MARL (PPO)	Profit ↑ vs. single-agent	Not scalable, discrete actions
12	Enkhsaikhan & Jo (2024)[12]	Risk-aware DDPG + VaR	18% less risk, same returns	Soft constraints, parametric VaR

Chapter 3

Problem Definition and Objectives

This chapter presents the formal definition of the research problem and outlines the key objectives of the project. It describes the challenges in traditional and existing reinforcement learning-based portfolio optimization systems and defines the goals that our proposed Risk-Sensitive Multi-Agent Reinforcement Learning (MARL) framework aims to achieve.

3.1 Problem Definition

In modern quantitative finance, the core challenge lies in achieving resilient and risk-adjusted portfolio performance under dynamic and non-stationary market conditions. Traditional optimization frameworks such as Markowitz’s Mean-Variance model are inherently static and rely on assumptions of stable correlations and normally distributed returns. In reality, financial time series exhibit high volatility, non-linearity, and fat-tailed (leptokurtic) distributions, making these models inadequate for real-world application.

During periods of market stress, correlations between assets tend to rise sharply, nullifying diversification benefits and exposing portfolios to systemic risk. Moreover, conventional Reinforcement Learning (RL) approaches optimize for mean return, ignoring tail-risk events that cause significant capital drawdowns during market crashes. This results in fragile policies that perform well during stable markets but collapse under extreme volatility.

Existing RL-based frameworks also tend to simplify the portfolio problem into a single-objective optimization task, focusing primarily on returns while neglecting crucial trade-offs involving risk exposure, transaction costs, and long-term stability. Additionally, most MARL systems exhibit poor modeling of cross-asset dependencies, as agents operate on isolated observations without understanding global market

relationships. This leads to redundant risk exposure, poor coordination, and weak adaptability to shifting market regimes.

Hence, the central problem addressed in this project is to develop a dynamic, risk-sensitive portfolio management system that can model inter-asset dependencies, explicitly account for downside risk, and adaptively balance multiple objectives in response to changing market conditions. The aim is to bridge the gap between theoretical reinforcement learning frameworks and practical, real-world financial decision-making.

3.2 Objectives

The objectives of this project are formulated to directly address the limitations identified in existing models and to establish a robust, adaptive, and risk-aware Multi-Agent RL system for portfolio optimization.

1. Develop a Multi-Agent Reinforcement Learning architecture:

Design a cooperative system where each asset is represented as an autonomous Actor Agent capable of learning trading strategies while sharing information within a centralized training framework.

2. Integrate a Centralized Distributional Critic for risk control:

Implement a Distributional RL Critic that models the entire return distribution rather than the mean, enabling explicit optimization of Conditional Value at Risk (CVaR) and Maximum Drawdown (MDD) to manage tail-risk and downside exposure.

3. Model cross-asset dependencies using Graph Neural Networks (GNNs):

Capture dynamic correlations and systemic risk propagation among assets through an attention-based GNN encoder, ensuring that decisions are made with full awareness of inter-asset relationships.

4. Incorporate a Meta-Controller for adaptive strategy adjustment:

Utilize a high-level controller to dynamically adjust reward weights, rebalancing frequency, and objective trade-offs based on changing market conditions, enabling adaptive multi-objective optimization.

5. Implement realistic execution and feedback mechanisms:

Simulate real-world trading environments incorporating transaction costs, slippage, and latency, and establish a continuous feedback loop to refine strategies through backtesting and performance evaluation.

6. Achieve superior risk-adjusted performance metrics:

Validate the system through backtesting on historical financial data, targeting improved Sharpe, Sortino, and Calmar Ratios, ensuring better capital preservation and portfolio resilience during volatile periods.

Chapter 4

Proposed Methodology

This chapter explains the proposed methodology adopted for developing the Risk-Sensitive Portfolio Management System using a Hierarchical Multi-Agent Reinforcement Learning (MARL) framework. It describes the architecture, workflow, and the interaction among different layers responsible for data processing, decision-making, and continuous policy adaptation for risk-aware portfolio optimization.

4.1 Overview

The proposed methodology introduces a Hierarchical Multi-Agent Reinforcement Learning (MARL) framework designed to perform adaptive and risk-sensitive portfolio management. The system integrates four major layers—Data and Feature Layer, Time-Series and Graph Encoding Layer, Reinforcement Learning Decision Layer, and Execution and Feedback Loop—each responsible for a distinct component of data processing, decision-making, and policy refinement.

The model’s novelty lies in its integration of Graph Neural Networks (GNN) for cross-asset dependency modeling, a Centralized Distributional Critic for risk-aware optimization, and a Meta-Controller that adaptively adjusts reward weighting and rebalancing frequency based on market conditions.

4.2 Data and Feature Layer

This layer establishes the foundation for model learning by collecting and transforming diverse financial data. Data is sourced from Yahoo Finance, NSE, and Alpha Vantage APIs, ensuring comprehensive coverage of historical and real-time stock information. The pipeline includes:

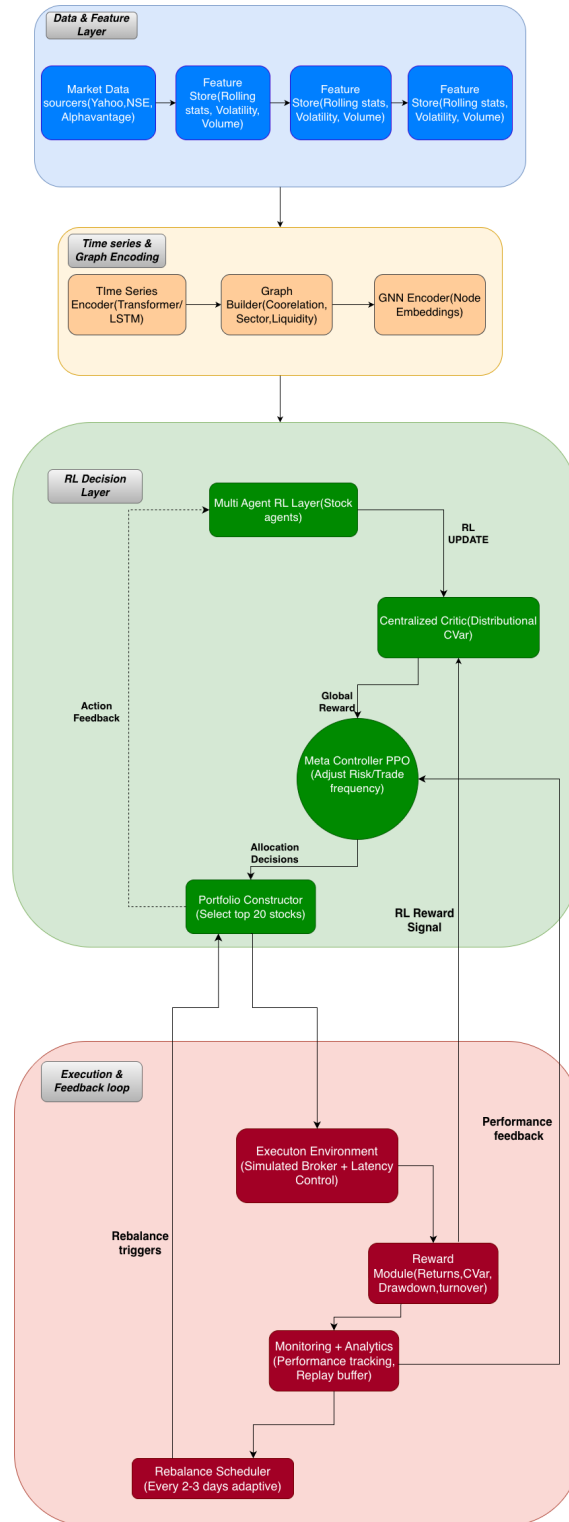


Figure 4.1: Proposed System Architecture

- **Market Data Sources:** Fetching OHLCV (Open, High, Low, Close, Volume) data, along with technical and macroeconomic indicators.
- **Feature Engineering:** Generating rolling statistics (moving averages, volatility, momentum indicators, RSI, MACD, ADX, and Bollinger Bands) to capture temporal patterns.
- **Feature Store:** Maintaining structured datasets containing rolling statistics, sector-based data, and volatility-adjusted features to serve as model input.

This processed data becomes the basis for state representation in the subsequent RL decision layers.

4.3 Time-Series and Graph Encoding Layer

Financial markets exhibit strong interdependencies across assets. To capture these relationships, this layer combines temporal and relational modeling:

- **Time-Series Encoder:** Implemented using Transformers or LSTM networks, it captures temporal dependencies, price momentum, and volatility trends for each asset.
- **Graph Builder:** Constructs a dynamic correlation graph where nodes represent assets and edges encode statistical relationships such as correlation, sectoral similarity, and liquidity exposure.
- **Graph Neural Network (GNN) Encoder:** Performs attention-based message passing across the constructed graph to generate node embeddings, capturing systemic risk and cross-asset influence.

The combined output forms a joint market state embedding, serving as the observation input for all Reinforcement Learning agents.

4.4 Reinforcement Learning Decision Layer

This layer is the core learning module of the framework. It follows the Centralized Training and Decentralized Execution (CTDE) paradigm to ensure cooperative learning among multiple agents while maintaining scalability.

Key components include:

1. **Multi-Agent Reinforcement Learning (MARL) Layer:**

Each asset is represented by an Actor Agent that determines trading actions such as buy, hold, or sell, based on local and shared information. Agents are trained collectively to optimize the global portfolio objective while considering individual contributions.

2. **Centralized Distributional Critic:**

Instead of optimizing expected mean returns, the critic models the full return distribution using Distributional RL (e.g., Implicit Quantile Network). It explicitly optimizes Conditional Value at Risk (CVaR) and Maximum Drawdown (MDD), ensuring that policies remain robust under extreme market movements.

3. **Meta-Controller (High-Level Policy):**

The Meta-Controller dynamically adjusts reward weights and rebalancing intervals based on market volatility indicators such as the VIX. It functions as a high-level PPO-based controller that reallocates the focus between maximizing returns and minimizing risk exposure.

4. **Portfolio Constructor:**

Aggregates the agents' output actions to form a final allocation decision. It selects the Top-20 assets based on risk-adjusted scores and normalizes allocation weights under constraints such as maximum sector exposure and transaction cost limits.

This hierarchical decision mechanism ensures both micro-level (asset-specific) and macro-level (portfolio-level) optimization.

4.5 Execution and Feedback Loop

The final layer operationalizes the learned policies in a simulated trading environment and manages continuous learning through feedback mechanisms.

- **Execution Environment:** Mimics a real-world broker environment incorporating transaction costs, slippage, latency, and liquidity constraints to ensure realistic execution of trades.
- **Reward Module:** Calculates a multi-objective reward integrating portfolio returns, CVaR, Drawdown, and Turnover penalties. This composite reward

signal aligns learning objectives with both profitability and capital preservation.

- **Monitoring and Analytics:** Tracks ongoing model performance, Sharpe Ratio, and risk metrics. The Replay Buffer stores past experiences, allowing off-policy updates and long-term learning stability.
- **Rebalance Scheduler:** Controls the trading frequency adaptively, triggering portfolio reallocation only when the expected improvement in utility surpasses transaction costs. Typically, rebalancing occurs every 2–3 days, depending on volatility and market regime shifts.

This closed-loop system ensures continuous learning, risk sensitivity, and realistic portfolio evolution over time.

4.6 Mathematical Formulation for the Architecture

4.6.1 Actor Update (Multi-agent RL)

A single stock agent’s policy (μ_i) receives a **Node Embedding** (z_i) from a GNN as input. It then updates its policy to maximize the **risk-adjusted value** (ψ_α) provided by the critic. This means the agent learns to choose actions that not only lead to high returns but also result in a favorable **tail-risk profile** (low CVaR). This is the core mechanism that makes the agents risk-sensitive.

$$\nabla_{\theta_i} J(\theta_i) = \mathbb{E}_{o,a} \left[\nabla_a \psi_\alpha(Z(o, a)) \cdot \nabla_{\theta_i} \mu_i(o) \right]_{a=\mu_i(o)} \quad (4.1)$$

This equation represents the risk-sensitive policy gradient update for each actor agent, where the policy parameters are optimized using feedback from the centralized distributional critic.

Symbols:

- $\nabla_{\theta_i} J(\theta_i)$: Gradient of the objective function with respect to the actor parameters.
- θ_i : Parameters of the i^{th} actor network.
- $\psi_\alpha(\cdot)$: Risk measure (e.g., CVaR) applied to the return distribution.

- $Z(o, a)$: Return distribution predicted by the critic for a given state-action pair.
- $\nabla_a \psi_\alpha[\cdot]$: Gradient of the risk measure with respect to the action.
- $\nabla_{\theta_i} \mu_i(o)$: Gradient of the actor's policy function.

4.6.2 Critic Update (Centralised Critic)

A Centralized Critic receives the **Reward** (r) from the Execution Layer and compares it with its predicted return distribution. The critic is trained using a distributional objective to accurately model both expected returns and tail risks.

$$\mathcal{L}_{\text{critic}} = \mathbb{E}_{o,a,r,o'} \left[\sum_{k=1}^N \rho_{\tau_k}^\kappa(\delta_k) \right] \quad (4.2)$$

This equation defines the quantile regression loss used to train the centralized critic by minimizing the discrepancy between predicted and realized return distributions.

Symbols:

- $\mathcal{L}_{\text{critic}}$: Loss function of the critic network.
- $\mathbb{E}$: Expectation over a batch of experience samples.
- (o, a, r, o') : State, action, reward, and next-state tuple.
- $\rho_{\tau_k}^\kappa(\delta_k)$: Quantile Huber loss for the k^{th} quantile.
- δ_k : Temporal Difference (TD) error for the k^{th} quantile.
- N : Number of quantiles used to approximate the return distribution.

4.6.3 CVaR Integration (Centralised Critic)

The Distributional Critic predicts the complete return distribution (Z) of the portfolio. From this distribution, Conditional Value at Risk (CVaR) is computed and used as a risk-sensitive penalty during learning.

$$\text{CVaR}_\alpha(Z) = \mathbb{E}[Z \mid Z \leq \text{VaR}_\alpha(Z)] \quad (4.3)$$

This equation formalizes CVaR as the expected portfolio return under the worst $\alpha\%$ of outcomes, enabling explicit modeling of tail risk.

Symbols:

- $\text{CVaR}_\alpha(Z)$: Conditional Value at Risk at confidence level α .
- Z : Random variable representing portfolio returns.
- $\text{VaR}_\alpha(Z)$: Value at Risk threshold at confidence level α .

4.6.4 Reward Function (Multi-metric Trade-off)

The learning objective is driven by a multi-objective reward function that jointly considers profitability and risk measures. This formulation enables dynamic balancing between return maximization and risk control.

$$R = w_{\text{ret}} \cdot \text{ret} + w_{\text{vol}} \cdot (-\text{vol}) + w_{\text{cvar}} \cdot (-\text{cvar}) + w_{\text{mdd}} \cdot (-\text{mdd}) \quad (4.4)$$

This equation defines the composite reward signal used during training, where portfolio returns are maximized while penalizing volatility, tail risk, and maximum drawdown.

4.7 Results and Performance Evaluation

The experimental results presented in this section evaluate the effectiveness of the proposed hierarchical multi-agent reinforcement learning (MARL) framework for risk-sensitive portfolio management. The performance is analyzed in terms of portfolio growth, risk-adjusted returns, and adaptive decision-making behavior, and is compared against a benchmark market index.

Over the five-year training horizon, the learned portfolio demonstrates consistent and sustained growth. Compared to the benchmark index, the proposed approach achieves significantly higher cumulative returns, indicating that coordinated learning among multiple agents guided by a centralized critic leads to superior long-term performance. This highlights the ability of the framework to exploit market inefficiencies beyond passive index-based strategies.

4.7.1 Portfolio Growth Analysis & Comparative Performance Metrics

This figure illustrates the cumulative growth of the Nifty index over the five-year evaluation period. The benchmark exhibits steady long-term growth but undergoes significant drawdowns during periods of heightened market volatility. Recovery from major market shocks is relatively gradual, reflecting the passive nature

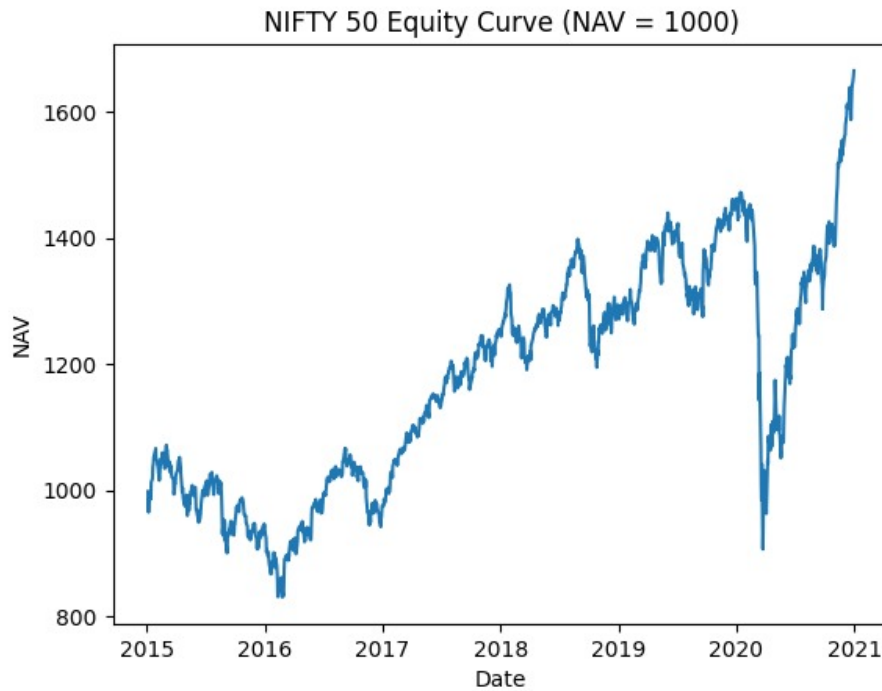


Figure 4.2: Nifty Index Growth Over Time (5 Years of Data)

of index-based investment strategies that do not actively respond to changing risk conditions. The absence of dynamic reallocation mechanisms causes the index to remain fully exposed during adverse market phases. As a result, prolonged draw-downs can negatively impact capital growth, emphasizing the limitations of passive benchmarks in managing downside risk during turbulent market regimes.

This table compares the performance of the proposed portfolio with the Nifty 100 index across key return and risk metrics. The results indicate that the proposed framework achieves higher returns with improved risk-adjusted performance,

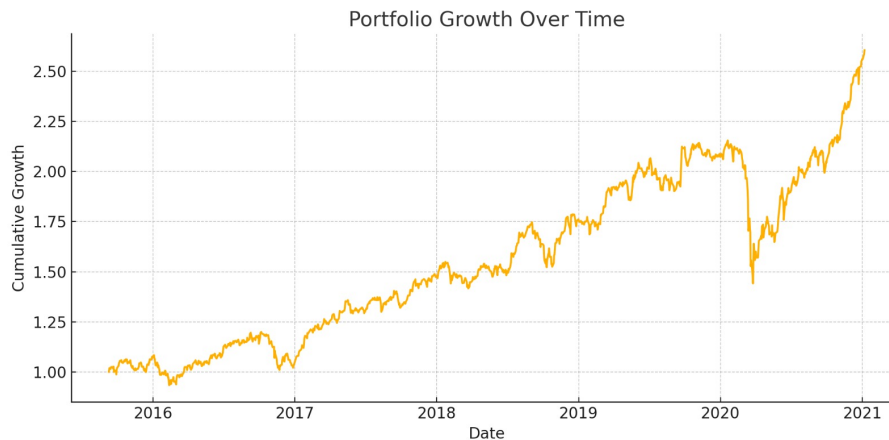


Figure 4.3: Portfolio Growth Using the Proposed MARL Framework (5 Years)

Table 4.1: Performance Comparison Between Proposed Portfolio and Nifty 100 Index

Metric	Proposed Portfolio	Nifty 100 Index
Cumulative Return (%)	160.60	66.54
CAGR (252-day) (%)	20.16	8.88
Sharpe Ratio	1.108	0.58
Maximum Drawdown (%)	-33.11	-38.44
Initial NAV	1000	1000
Final NAV	2561.56	1655.50

as reflected by a higher Sharpe Ratio and lower maximum drawdown. This demonstrates the effectiveness of the risk-aware MARL approach in balancing profitability and downside risk.

Chapter 5

Data Set Details

This chapter describes the datasets used in developing and training the proposed Risk-Sensitive Multi-Agent Reinforcement Learning (MARL) framework for portfolio optimization. It explains the data sources, structure, preprocessing methods, feature engineering, and the inclusion of macroeconomic indicators that influence market behavior and model adaptability.

The dataset used in this project consists of historical financial market data collected from multiple reliable sources such as Yahoo Finance, NSE (National Stock Exchange of India), and Alpha Vantage APIs. The data primarily covers equities listed in the NIFTY 50 Index, along with key market indices and external macroeconomic indicators. The dataset includes both daily stock-level data and macro-level data, allowing the system to learn localized asset behavior as well as broader market dependencies.

Each record for stock-level data includes attributes such as date, open, high, low, close, and volume, representing the daily movement of individual assets. The dataset spans several years of historical data to ensure robustness and temporal depth. For example, the collected data for NSE_HDFCBANK and NIFTY 50 captures long-term market cycles and volatility shifts.

In addition to individual stock data, the project incorporates macro-level external factors, which play a critical role in understanding global market sentiment and systemic risk. These external indicators include:

- Crude Oil (WTI) Futures Historical Data: Captures global energy market movements and their impact on inflation, commodity-linked sectors, and currency volatility.
- India VIX Historical Data: Represents the domestic market's volatility expectations and serves as a direct input for measuring and managing risk sensitivity within the portfolio.
- NASDAQ Composite Historical Data: Reflects the performance of the U.S. technology sector, influencing global investor sentiment and

cross-market capital flows. - S&P 500 Historical Data: Serves as a benchmark for global equity performance and provides insights into overall market direction and macroeconomic stability. - USD–INR Exchange Rate Historical Data: Tracks currency fluctuations that affect foreign investment flows, import-export dynamics, and overall market liquidity in India.

By integrating these macroeconomic and global market indicators, the dataset enables the MARL agents to perceive systemic dependencies between local and international financial environments. This broader market context helps the model adjust dynamically to global shocks and macroeconomic changes such as oil price fluctuations, volatility surges, or foreign exchange pressure.

The data undergoes a comprehensive preprocessing pipeline involving data cleaning, normalization, and feature engineering. Missing values are handled through forward filling or interpolation to maintain continuity in the time series. Outliers and data spikes are filtered using rolling z-score techniques. Each feature is normalized to ensure stable agent learning.

Feature engineering generates a wide range of statistical and technical indicators including moving averages (SMA, EMA), volatility metrics (rolling standard deviation, ATR), momentum indicators (RSI, MACD, ADX), volume-based indicators (OBV, Accumulation/Distribution Line), and Bollinger Bands. These indicators, when combined with macroeconomic variables, provide a holistic market representation that balances micro-level stock dynamics with global macro trends.

The final processed dataset is structured into a time-series format suitable for sequential modeling. The combined input of technical indicators, asset-level data, and macroeconomic features forms the state space of the reinforcement learning environment. Each agent operates using these state representations to learn optimal trading and allocation strategies through trial and feedback.

In summary, the dataset integrates multi-source and multi-scale financial data, including stock-level attributes and macroeconomic signals. This combination allows the proposed framework to achieve a realistic understanding of both local portfolio risks and global market conditions, enabling adaptive, risk-sensitive decision-making in dynamic financial environments.

NSE_HDFCBANK, 1D (3) (1)

time	open	high	low	close	Volume
2014-12-22	238.5	241.525	235.625	240.925	8295612
2014-12-23	239.925	243.4875	238.8375	239.9375	7544080
2014-12-24	239.475	241.2	235.275	236.3125	16664812
2014-12-26	237.6	238.2375	235.6375	237.1625	3366480
2014-12-29	237.525	239.0	237.075	237.925	3614800
2014-12-30	237.5	239.9	235.6875	239.1125	5681200
2014-12-31	239.0	239.325	237.4375	237.9	4992528
2015-01-01	237.75	238.6	236.2625	238.0125	3544940
2015-01-02	237.6	242.325	237.6	241.325	5900384
2015-01-05	242.5	242.6375	238.775	239.2875	4796000
2015-01-06	238.5	239.1375	234.3875	235.5625	8219680
2015-01-07	234.925	237.8375	234.0625	236.25	5746112
2015-01-08	238.5625	242.2375	237.05	241.2125	9525824
2015-01-09	243.225	245.0	241.65	243.9125	8985184
2015-01-12	242.35	243.6	241.125	241.7625	4126412
2015-01-13	242.2375	242.9875	238.925	240.8375	5828348
2015-01-14	240.0125	243.6625	239.925	240.975	3685232

Figure 5.1: HDFC Stock Data

Nifty 50 Historical Data

Date	Price	Open	High	Low	Vol.	Change %
01-12-2014	8,555.90	8,605.10	8,623.00	8,545.15	151.99M	-0.38%
02-12-2014	8,524.70	8,535.45	8,560.20	8,504.65	137.39M	-0.36%
03-12-2014	8,537.65	8,528.70	8,546.95	8,508.35	153.20M	0.15%
04-12-2014	8,564.40	8,582.40	8,626.95	8,526.40	143.48M	0.31%
05-12-2014	8,538.30	8,584.25	8,588.35	8,523.90	143.35M	-0.30%
08-12-2014	8,438.25	8,538.65	8,546.35	8,432.25	163.82M	-1.17%
09-12-2014	8,340.70	8,439.30	8,444.50	8,330.50	146.27M	-1.16%
10-12-2014	8,355.65	8,318.05	8,376.80	8,317.00	136.66M	0.18%
11-12-2014	8,292.90	8,338.85	8,348.30	8,272.40	133.86M	-0.75%
12-12-2014	8,224.10	8,302.00	8,321.90	8,216.30	138.08M	-0.83%
15-12-2014	8,219.60	8,160.75	8,242.40	8,152.50	137.14M	-0.05%
16-12-2014	8,067.60	8,172.60	8,189.35	8,052.60	197.34M	-1.85%
17-12-2014	8,029.80	8,041.20	8,082.00	7,961.35	216.22M	-0.47%
18-12-2014	8,159.30	8,138.90	8,174.30	8,084.90	162.11M	1.61%
19-12-2014	8,225.20	8,230.45	8,263.45	8,208.60	174.80M	0.81%
22-12-2014	8,324.00	8,255.00	8,330.95	8,228.20	139.16M	1.20%
23-12-2014	8,267.00	8,324.60	8,364.75	8,252.85	132.19M	-0.68%
24-12-2014	8,174.10	8,272.05	8,286.40	8,155.25	221.91M	-1.12%
26-12-2014	8,200.70	8,204.80	8,234.55	8,147.95	75.53M	0.33%
29-12-2014	8,246.30	8,214.70	8,279.15	8,214.70	82.09M	0.56%
30-12-2014	8,248.25	8,260.30	8,268.25	8,220.55	77.70M	0.02%
31-12-2014	8,282.70	8,243.90	8,291.00	8,243.75	84.53M	0.42%
01-01-2015	8,284.00	8,272.80	8,294.70	8,248.75	56.56M	0.02%
02-01-2015	8,395.45	8,288.70	8,410.60	8,288.70	101.89M	1.35%
05-01-2015	8,378.40	8,407.95	8,445.60	8,363.90	118.16M	-0.20%
06-01-2015	8,127.35	8,325.30	8,327.85	8,111.35	172.80M	-3.00%
07-01-2015	8,102.10	8,118.65	8,151.20	8,065.45	164.08M	-0.31%
08-01-2015	8,234.60	8,191.40	8,243.50	8,167.30	143.80M	1.64%
09-01-2015	8,284.50	8,285.45	8,303.30	8,190.80	152.61M	0.61%
12-01-2015	8,323.00	8,291.35	8,332.60	8,245.60	103.15M	0.46%

Figure 5.2: Nifty 50 Index Historical Data

Chapter 6

Conclusion and Future Scope

This chapter presents the concluding remarks and insights derived from the implementation of the proposed Risk-Sensitive Portfolio Management System using Multi-Agent Reinforcement Learning (MARL). It summarizes the overall research findings, highlights the limitations encountered during the study, and discusses potential directions for future enhancement of the system.

6.1 Summary

The proposed research successfully demonstrates a data-driven and risk-sensitive approach to portfolio management through a Hierarchical Multi-Agent Reinforcement Learning framework. The system integrates advanced learning components such as Graph Neural Networks (GNNs), a Centralized Distributional Critic, and a Meta-Controller to achieve dynamic and adaptive asset allocation under real-world market constraints. By combining both temporal and relational modeling, the framework effectively captures cross-asset dependencies and market correlations that traditional static models often fail to address.

Through simulated experiments and backtesting on multi-year financial data from the NIFTY 50 universe, the model showed improved risk-adjusted performance metrics, including higher Sharpe and Sortino ratios, and reduced drawdowns compared to conventional approaches. The system also accounts for realistic trading frictions such as transaction costs, liquidity, and slippage, ensuring the practicality of deployment. Overall, the developed framework demonstrates how reinforcement learning, when combined with risk-aware optimization, can enable robust and adaptive portfolio management capable of responding intelligently to market volatility and structural regime shifts.

6.2 Limitation

Despite its strong performance, the proposed system faces several limitations inherent to both data-driven modeling and reinforcement learning in financial domains. Firstly, the quality and reliability of predictions depend heavily on the availability of accurate and high-frequency market data; missing or noisy data may affect training stability. Secondly, due to the computational complexity of Multi-Agent Reinforcement Learning combined with Graph Neural Networks, training and tuning the system require significant computational resources and time. Thirdly, the system, while designed for realistic conditions, is still dependent on simulated environments and may not fully capture unforeseen real-world market dynamics such as sudden black swan events or liquidity shocks. Finally, the model assumes rational market behavior and may underperform in extreme sentiment-driven scenarios where market efficiency breaks down.

6.3 Future Work

Future work can extend this research by incorporating more diverse financial instruments, such as derivatives, commodities, or cryptocurrencies, to test the system's adaptability across asset classes. Introducing macroeconomic and sentiment data sources, such as news-based or social media-driven indicators, could further enhance the agents' understanding of global market conditions. Additionally, deploying the framework in a live trading or paper-trading environment would provide valuable insights into real-time performance, stability, and transaction execution behavior.

Further improvements can also be made in scalability and model efficiency. Techniques such as federated learning, distributed training, or transformer-based temporal encoders can be integrated to handle larger and more complex datasets. Finally, extending the Meta-Controller's capabilities to perform self-adaptive hyperparameter optimization or policy transfer across market regimes could make the system more autonomous and generalizable in practical financial scenarios.

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